

Project – Customer Personality Analysis

Introduction:

In a competitive and customer-centric marketplace, understanding consumer behavior is essential for effective marketing strategies. The Customer Personality Analysis Dashboard was developed to explore purchasing patterns, campaign responses, demographics, and behavioral metrics using data from a retail marketing campaign. The dataset, `marketing_campaign.csv`, contains key insights into customer demographics, purchasing channels, product interests, income levels, and responses to marketing efforts. Visuals such as bar charts, donut charts, density plots, and histograms make it easy to identify customer trends and optimize future marketing and sales strategies. This comprehensive analysis bridges the gap between data and decisions by uncovering actionable insights from raw data.

Steps:

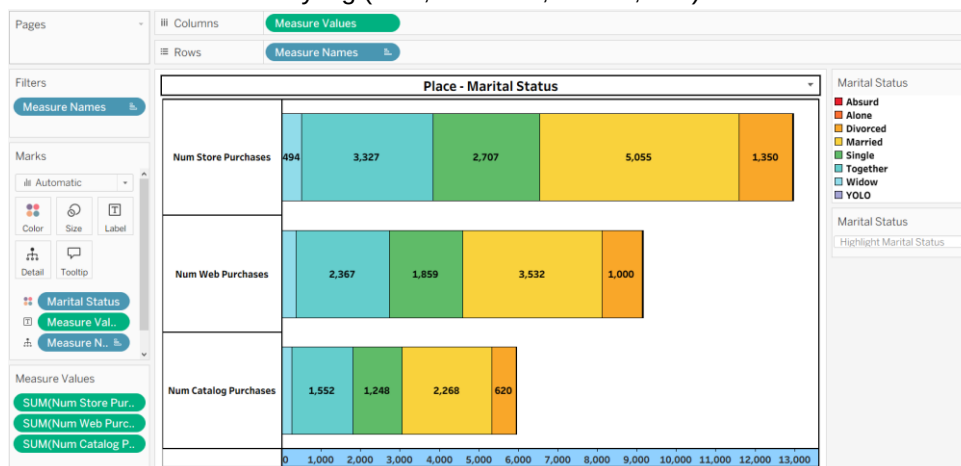
1. Load Data:

Loaded data by creating connection with 'marketing_campaign.csv'

2. Sheet 1: Place - Marital Status

Objective: Compare the number of purchases across web, catalog, and store based on marital status.

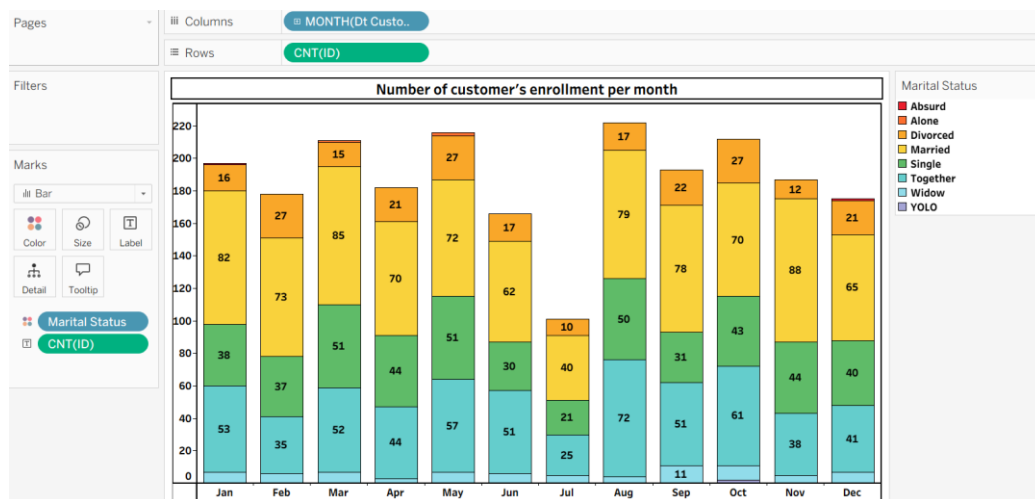
- 2.1. Dragged Measure Names to filters → Deselect All → Select Num Web Purchases, Num Web Purchases and Num Catalogue Purchases
- 2.2. Ctrl + Dragged Measure Names to Row
- 2.3. Dragged Measure Values to Column
- 2.4. Mark → Bar
- 2.5. Sorted to descending
- 2.6. Dragged Marital Status to Color and edited colors to Jewel Bright
- 2.7. Ctrl + Dragged Measure Values from Columns to Label
- 2.8. Formatted the sheet for styling (bold, centered, border, etc.).



3. Sheet 2: Number of customer's enrollment per month

Objective: Track customer onboarding trends monthly, segmented by marital status.

- 3.1. Duplicated the 'Marital Status' Sheet
- 3.2. Removed All values
- 3.3. Dragged 'Dt Customer' to Columns → Month
- 3.4. Dragged 'ID' to Rows → Count
- 3.5. Set Marks as Bar
- 3.6. Dragged 'Marital Status' to Color
- 3.7. Ctrl + CNT(ID) from Rows to Label



4. Sheet 3: Number web visits per month - Marital status

Objective: Analyze web engagement levels of customers, grouped by marital status.

4.1. Duplicated the Sheet

4.2. Removed Columns value and added 'Num Web Visits Month'

4.3. Edited Axis to set Tick Interval and removed Title

Edit Axis [Num Web Visits Month]

General

Tick Marks

Major Tick Marks

☐ Automatic

☒ Fixed

☐ None

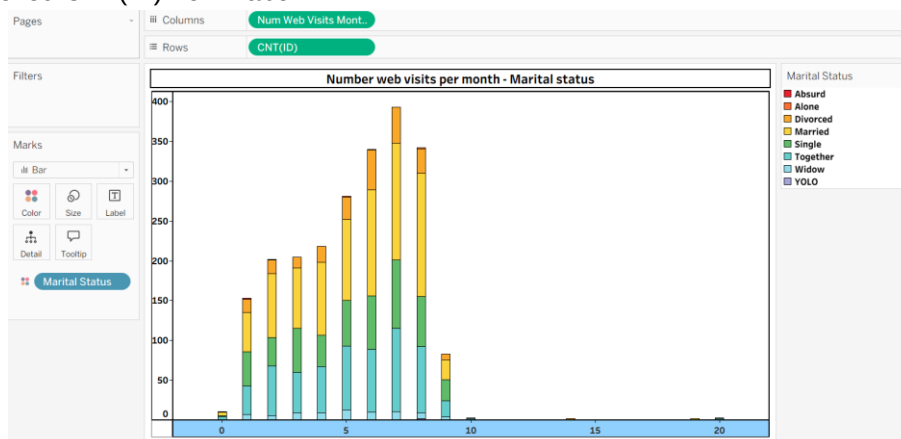
Tick origin

0

Tick interval

5

4.4. Removed CNT(ID) from Label



5. Sheet 4: Products – Marital Status

Objective: Identify product category preferences among customers based on marital status.

5.1. Duplicated the Sheet

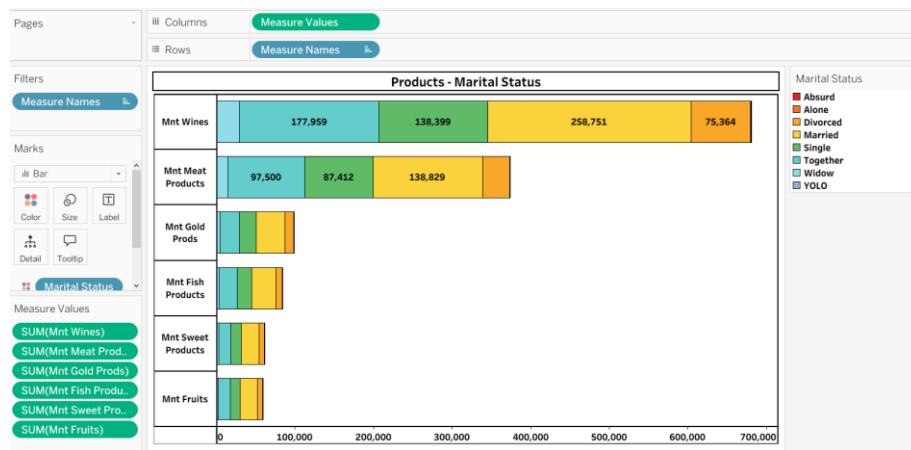
5.2. Removed Columns and Rows values

5.3. Ctrl + Dragged Measure Names to Filters and selected Products

5.4. Ctrl + Dragged Measure Names from Filters to Rows

5.5. Dragged Measure Values to Columns and Label

5.6. Edited Axis to removed Title



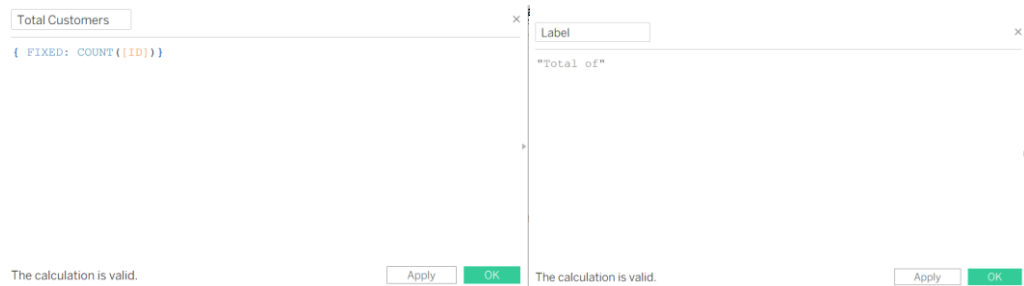
6. Sheet 5: Complaints

Objective: Visualize the proportion of customers who submitted complaints.

6.1. Duplicated the Sheet

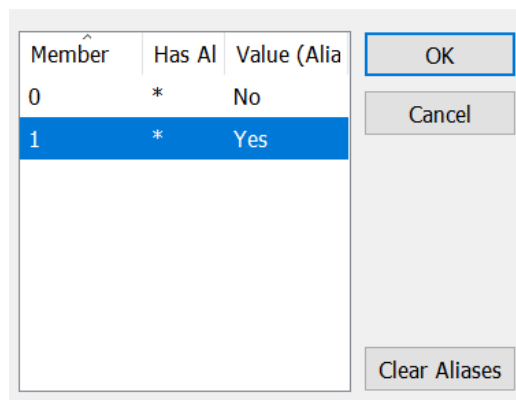
6.2. Removed all the values

6.3. Created two Calculated Fields:



6.4. Converted 'Complain' into dimension and edit aliases:

Edit Aliases [Complain]



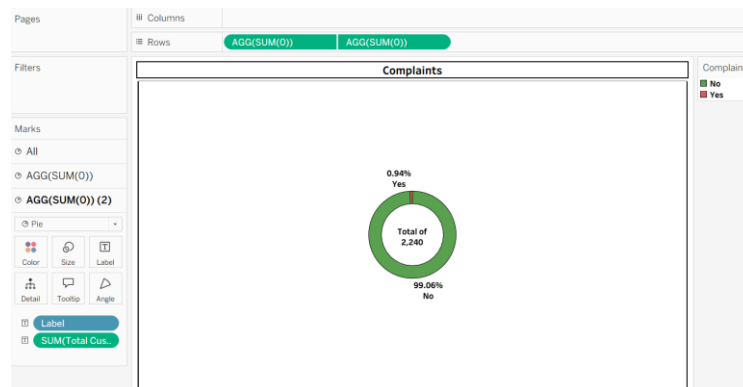
6.5. Dragged 'Complain' into Color

6.6. Set Marks as Pie

6.7. Dragged ID into Angle → Count

6.8. Label → Complain → Percent of Total; Label → CNT(ID)

6.9. Edited Color of 'Complain'. And Converted to Donut Chart



7. Sheet 6: First Campaign - Yes

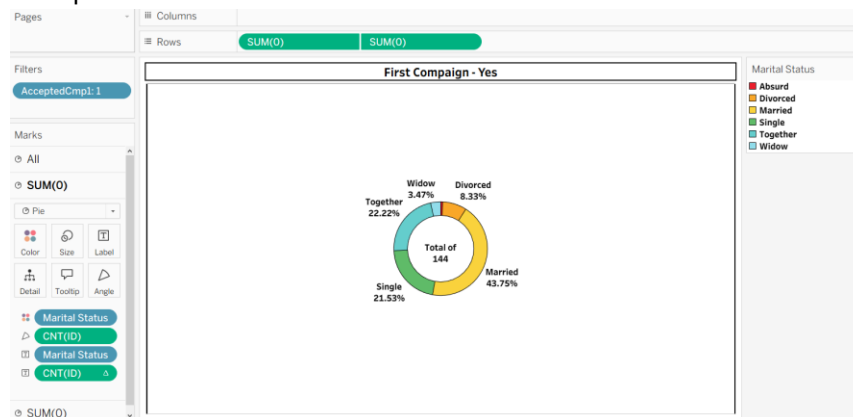
Objective: Understand 1st campaign performance and customer engagement, segmented by marital status.

7.1. Duplicated the Sheet

7.2. Converted 'AcceptedCmp1' into dimension and Dragged to Filters

7.3. Removed 'Total Customer' from 2nd Pie and added ID → Count

7.4. Removed 'Complain' from 1st Pie and added 'Marital Status' to Color

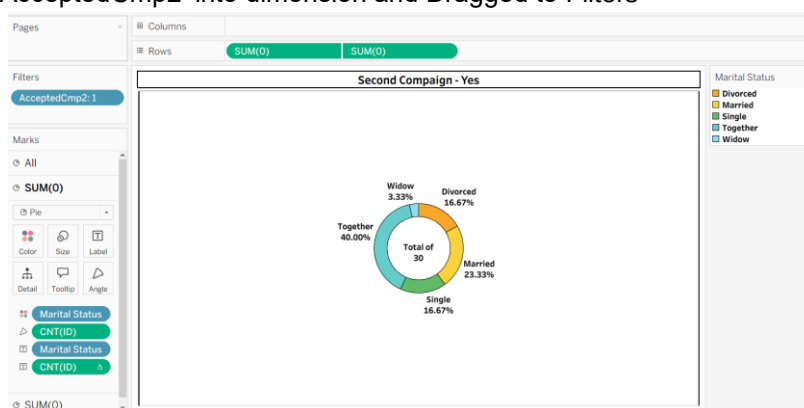


8. Sheet 7: Second Campaign - Yes

Objective: Understand 2nd campaign performance and customer engagement, segmented by marital status.

8.1. Duplicated the Sheet

8.2. Converted 'AcceptedCmp2' into dimension and Dragged to Filters



9. Sheet 8: Third Campaign - Yes

Objective: Understand 3rd campaign performance and customer engagement, segmented by marital status.

9.1. Duplicated the Sheet

9.2. Converted 'AcceptedCmp3' into dimension and Dragged to Filters

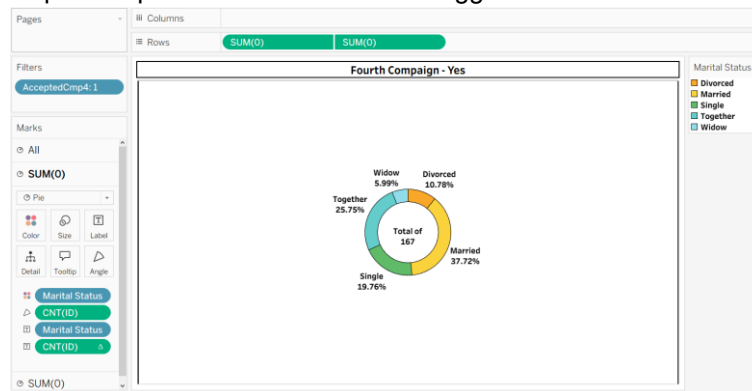


10. Sheet 9: Fourth Campaign - Yes

Objective: Understand 4th campaign performance and customer engagement, segmented by marital status.

10.1. Duplicated the Sheet

10.2. Converted 'AcceptedCmp4' into dimension and Dragged to Filters

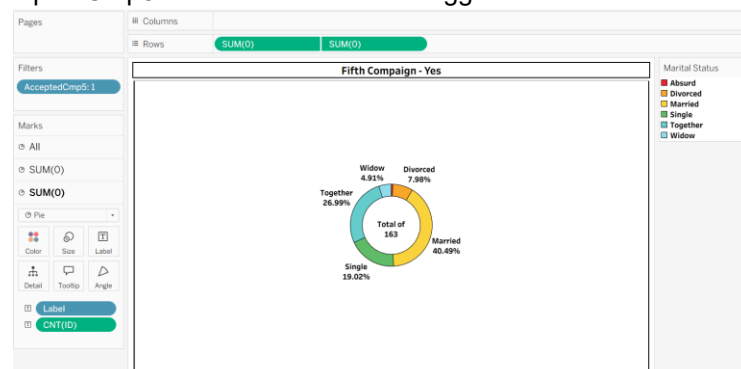


11. Sheet 10: Fifth Campaign - Yes

Objective: Understand 5th campaign performance and customer engagement, segmented by marital status.

11.1. Duplicated the Sheet

11.2. Converted 'AcceptedCmp5' into dimension and Dragged to Filters

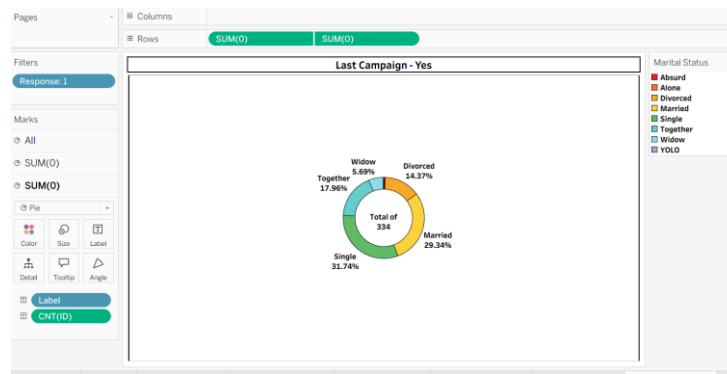


12. Sheet 11: Last Campaign - Yes

Objective: Understand last campaign performance and customer engagement, segmented by marital status.

12.1. Duplicated the Sheet

12.2. Dragged 'Response' to Filters and selected 1



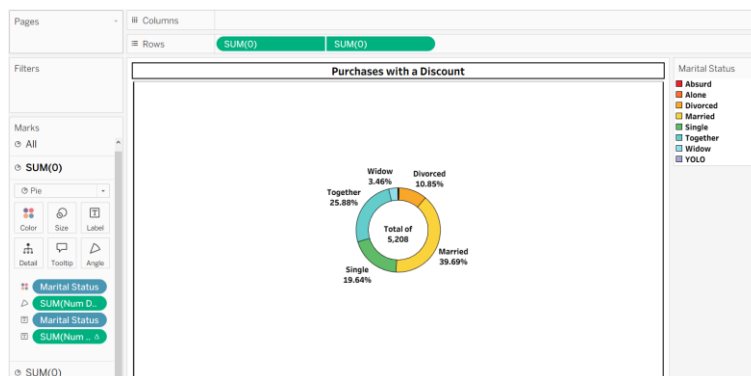
13. Sheet 12: Purchases with a Discount

Objective: Highlight the proportion of customers who made purchases using discounts, by marital status.

13.1. Duplicated the Sheet

13.2. Replaced all the ID values with 'Num Deal Purchases'

13.3. Removed the Filter



14. Sheet 13: Avg. Income - Marital Status

Objective: Compare average income levels among different marital groups.

14.1. Duplicated the Sheet

14.2. Removed All values

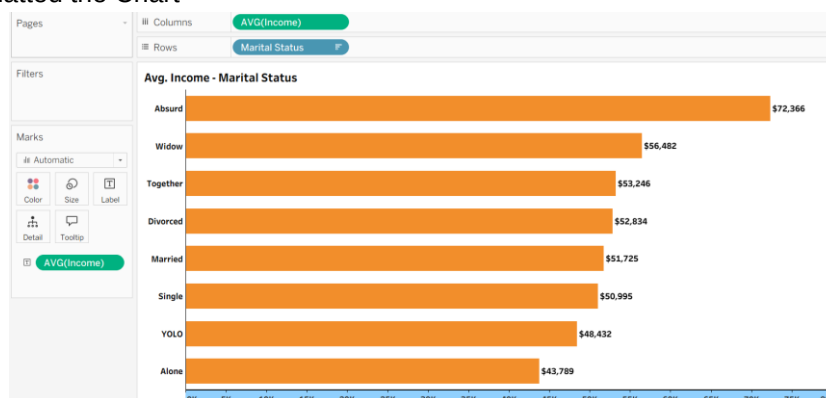
14.3. Dragged 'Income' to Columns → Average

14.4. Dragged 'Marital Status' to Rows

14.5. Set Marks as Bar

14.6. Ctrl + Dragged 'AVG(Income)' to Label

14.7. Formatted the Chart



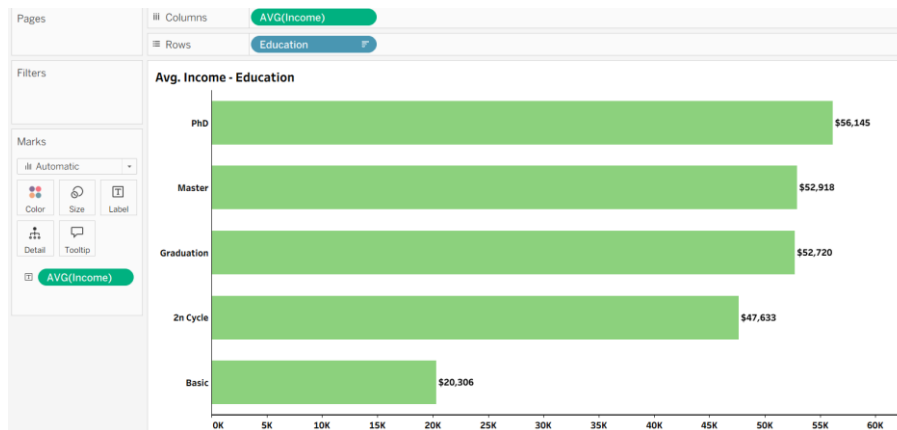
15. Sheet 14: Avg. Income - Marital Status

Objective: Understand how income varies by customer education level.

15.1. Duplicated the Sheet

15.2. Removed 'Marital Status' from Rows and Dragged 'Education'

15.3. Formatted the Chart



16. Sheet 15: People by Age and Marital Status

Objective: Analyze the age distribution of customers by marital status.

16.1. Duplicated the Sheet

16.2. Removed all values

16.3. Created a Calculated field:

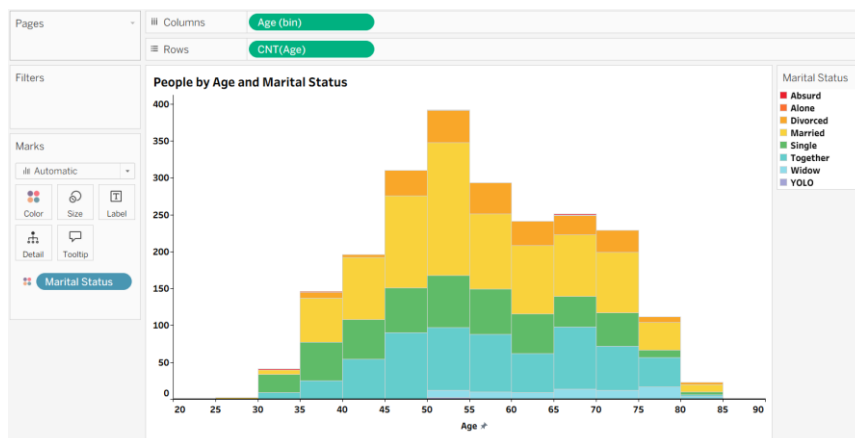


16.4. Double clicked in 'Age' → Show Me → Histogram

16.5. Edited the axis

16.6. Dragged 'Marital Status' to color

16.7. Formatted the Chart

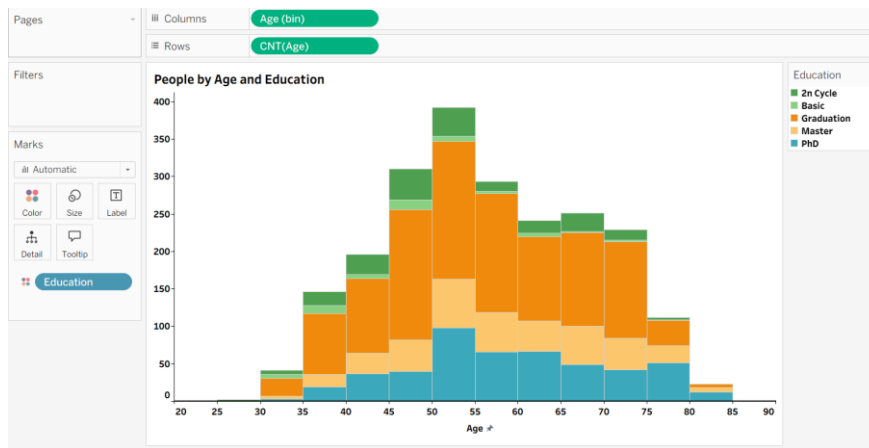


17. Sheet 16: People by Age and Education

Objective: Explore how age varies across different education levels.

17.1. Duplicated the Sheet

17.2. Removed 'Marital Status' from Color and added 'Education'



18. Sheet 17: Kids at Home - Marital Status

Objective: Visualize the distribution of households with children, segmented by marital status.

18.1. Duplicated the Sheet

18.2. Removed CNT(Age) from Rows and added Marital Status

18.3. Removed 'Marital Status' from Color and added Kidhome

18.4. Mark → Density

18.5. Formatted the Chart Color and Size

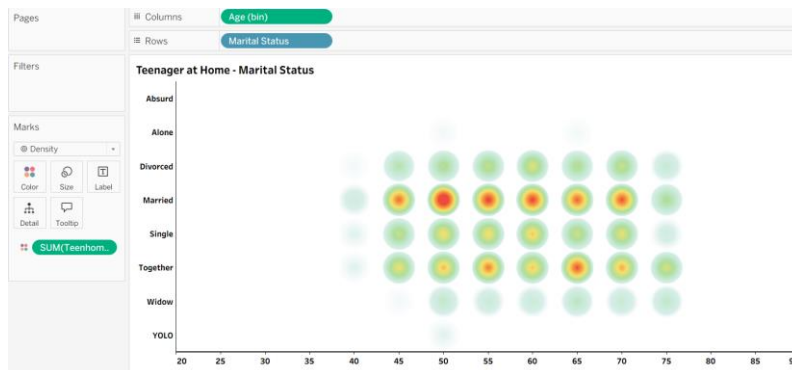


19. Sheet 18: Teenager at Home - Marital Status

Objective: Analyze the presence of teenagers in households by marital status.

19.1. Duplicated the Sheet

19.2. Removed 'Kidhome' from Color and added 'Teenhome'.



1. Dashboard

1.1. Set customize size 1850X1050

1.2. Added all the Charts

1.3. Apply final formatting such as removing gridlines, adding borders, and color coding for better visual appeal.



Key Insights:

- Store purchases outpace web and catalog channels, with married and together customers contributing the most.
- The highest product purchase volume is for wines, followed by meat products, again led by married customers.
- Enrollment trends peak in May and September, showing potential seasonal marketing opportunities.
- Complaints are minimal (only ~1%), indicating overall customer satisfaction.
- Campaign 1 and Last Campaign received the most positive responses, especially from married and together segments.
- Customers with higher education (Master's and PhD) report higher average income, making them key targets for premium offerings.
- Age groups 30–50 are the most prominent across all segments, suggesting mid-aged customers form the core demographic.
- Households with kids and teenagers are more common among married and together categories.

Conclusion:

The Customer Personality Analysis dashboard provides a comprehensive overview of consumer demographics, behaviors, and campaign interactions. By breaking down key KPIs across marital status, education, age, and product interests, businesses can fine-tune marketing strategies, personalize campaigns, and better align product offerings with customer preferences.

This data-driven approach enhances the company's ability to segment customers, maximize campaign ROI, and build stronger relationships with targeted user groups. The dashboard serves as a valuable decision-support tool for marketers, data analysts, and strategic planners alike.